

Case-based learning

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Lecturer

DIKIOHS, DUHS.

CBL # 1

Clinical scenario

A 68-year-old man is brought to the emergency department with sudden weakness of right side of his face & difficulty speaking. His family reports that he was normal until 2hrs ago.

On examination:

- ☐ He is conscious & understands spoken commands.
- ☐ There is weakness of right lower half of face.
- ☐ He has difficulty producing clear speech.
- ☐ He can raise both eye brows & close both eyes normally.
- ☐ His tongue deviates to the right when protruded.
- ☐ There is mild weakness of right upper limb.
- ☐ Sensory examination is normal.
- ☐ MRI shows an infarction involving left lateral frontal cortex, including region corresponding to the primary motor cortex & motor areas controlling the face.

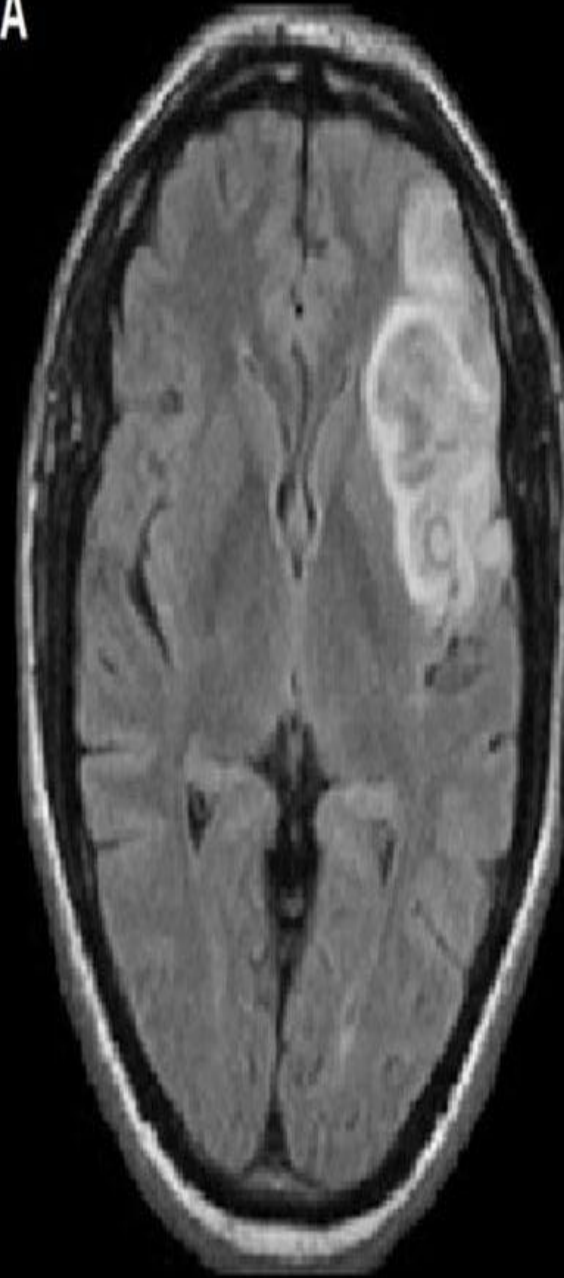
A 68-year-old man suddenly develops slurred speech and weakness of the right side of his face and arm while eating breakfast



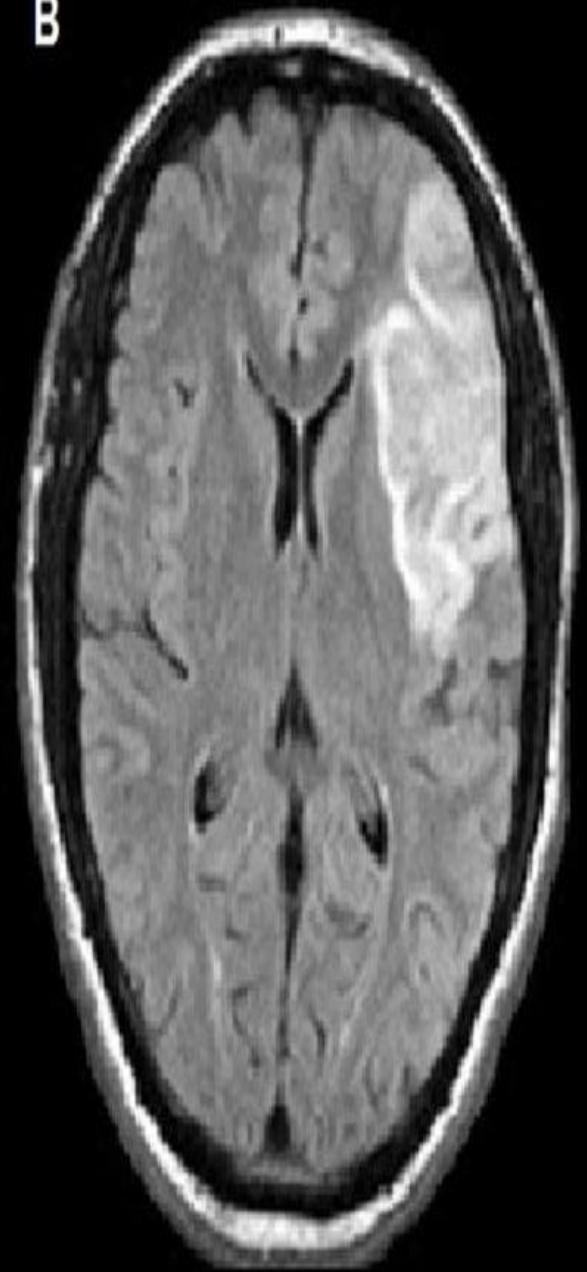
What is the most likely diagnosis?

- | | |
|--------------------------|---------------------------|
| 1. Bell's palsy | 2. Trigeminal nerve palsy |
| 3. Acute ischemic stroke | 4. Ramsay Hunt syndrome |

A



B



Questions

Q1. Which part of primary motor cortex controls movements of face, lips & tongue?

Q2. Why does a lesion in the left motor cortex produces weakness predominantly on the right side of the face?

Q3. Which cranial motor nucleus is particularly important in explaining pt's right lower facial weakness?

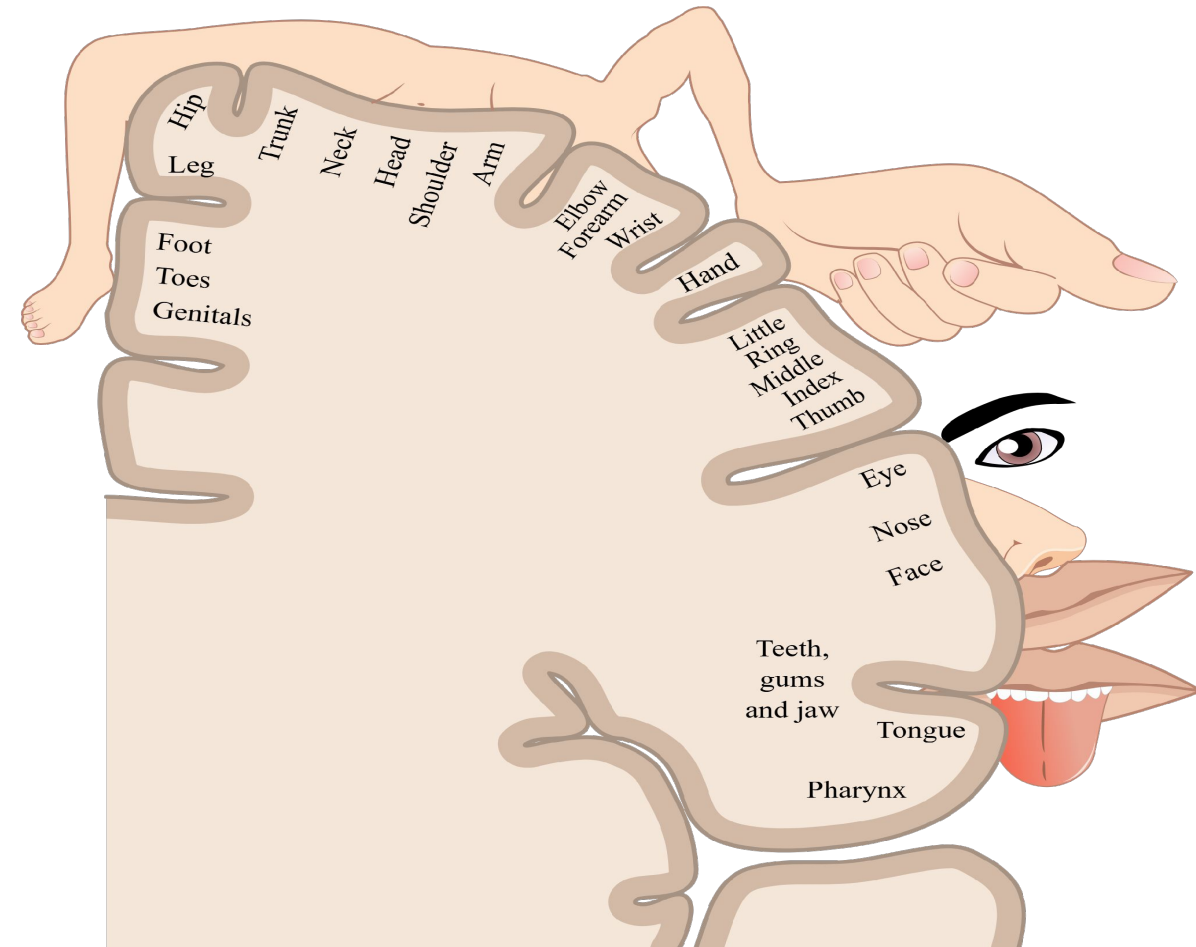
Q4. Why are the patient's forehead movements preserved despite weakness of right lower face?

Q5. The patient's tongue deviates to the right, which cranial nerve is involved?

Q6. Describe corticobulbar pathway.

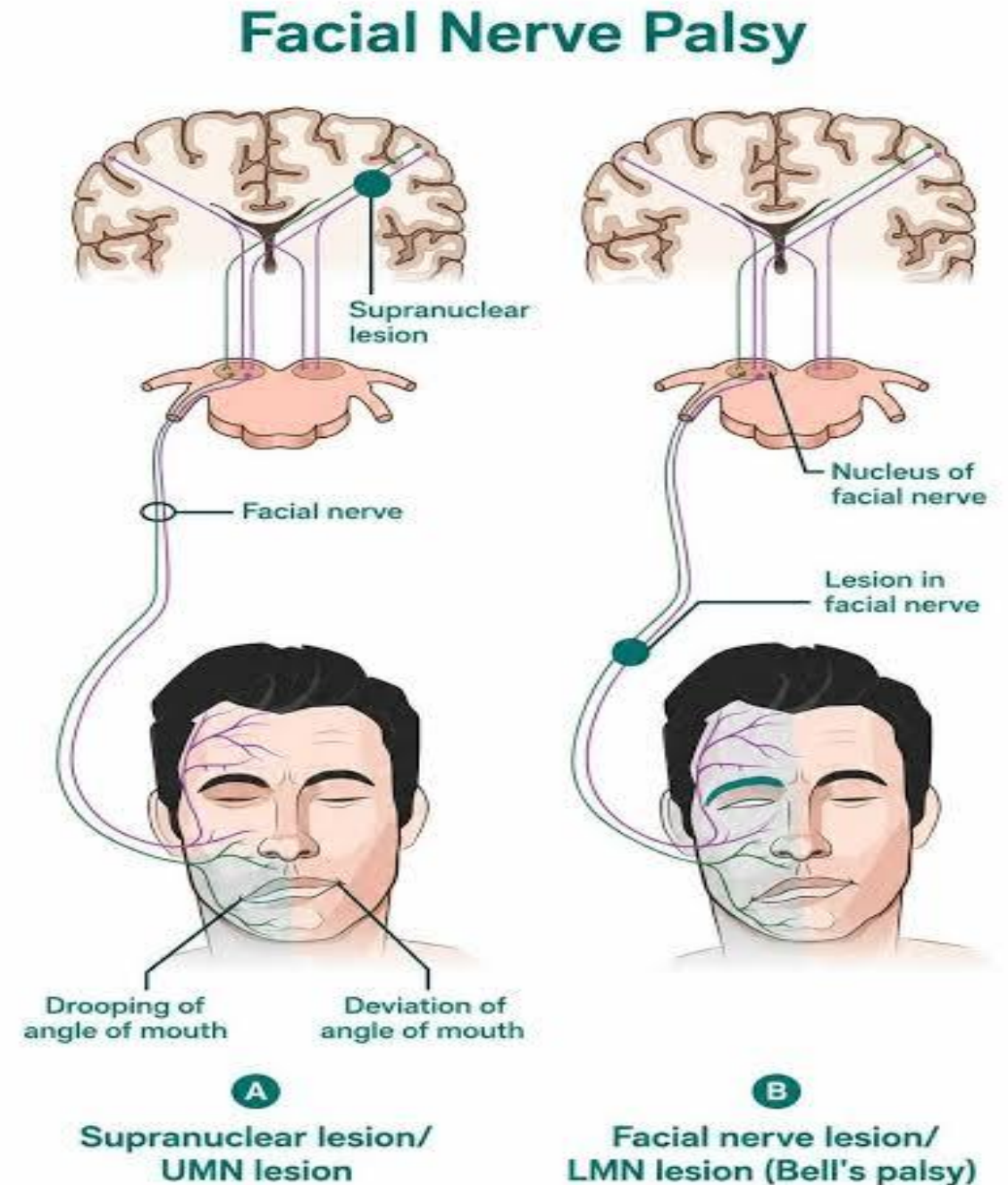
Answer # 1

The movements of face, lips & tongue are controlled by inferior lateral(inferolateral)region of primary motor cortex as the body is mapped in upside down(inverted) manner on motor homunculus with feet & legs mapped near the top of medial side while head, mouth, lips & tongue are mapped at the bottom of lateral side.



Answer # 2

Motor pathways that control voluntary facial movements cross over (decussate) from one side of the brain to stimulate cranial nerve nuclei of the opposite side.



Answer # 3

The right motor nucleus of CN-VII(facial nerve) is the key structure involved in right lower facial weakness.

Answer # 4

The neurons controlling right lower face (like smile & mouth) receives motor signals only from left motor cortex (***contralateral supply only***) whereas, neurons controlling forehead & eyes receives signals from both left & right motor cortex (***bilateral backup***) so if left motor cortex is damaged, right motor cortex still sends signals to both sides of forehead.

Therefore upper face (like eye-brow raising & eye closure) is usually spared or only mildly affected.

Answer # 5

The right hypoglossal nerve (CN XII) is involved in the deviation of tongue to the right side.

Answer # 6

Corticobulbar tract (also called corticonuclear tract) is a motor pathway that connects cerebral cortex to the motor nuclei of cranial nerves in the brainstem controlling voluntary movements of face, head & neck (like control of facial expression, mastication, swallowing, phonation, eye & tongue movements).

Scenario

A 20-year-old student accidentally touches a hot metal surface while working in the laboratory. She immediately withdraws her hand & experiences severe pain. However, after being assured by her teacher & focusing on the treatment, she reports that the pain has become less intense.

Later, she recalls that during a previous sport injury, she continued playing for several minutes despite having a significant injury & noticed surprisingly little pain until the game ended.

Related BCQs

BCQ # 1

What phenomenon explains why the perception of pain can decrease despite continued nociceptive input?

- a) Sensory adaptation
- b) Descending pain modulation
- c) Muscle spindle activation
- d) Proprioceptive inhibition
- e) Vestibular adaptation

BCQ # 2

Which structure is the important starting point of descending pain inhibitory system?

- a) Hippocampus
- b) Primary motor cortex
- c) Caudate nucleus
- d) Periaqueductal gray
- e) Cerebellar cortex

BCQ # 3

The PAG activates which major brainstem region involved in descending pain control?

- a) Lateral geniculate nucleus
- b) Inferior colliculus
- c) Rostral ventromedial medulla(RVM)
- d) Substantia nigra
- e) Vestibular nuclei

BCQ # 4

Which neurotransmitter is particularly important in descending inhibitory pathway from the brainstem to the spinal cord?

- a) Serotonin
- b) Histamine
- c) Glutamate only
- d) Substance P only
- e) Acetylcholine

BCQ # 5

Which neurotransmitter is released by inhibitory interneurons in the dorsal horn of spinal cord & contributes to analgesia?

- a) Dopamine
- b) Cortisol
- c) Histamine
- d) Endogenous opioids
- e) Acetylcholine

BCQ # 6

Interneurons in the dorsal horn release endogenous opioids. What is the important consequence?

- a) Increased release of substance P
- b) Increased release of glutamate
- c) Inhibition of nociceptive transmission
- d) Activation of motor neurons
- e) Increased transmission to higher centers

BCQ # 7

Which statement best describes the relationship between pain perception & nociceptive input?

- a) Pain intensity always directly reflects tissue damage
- b) Pain cannot be modified by higher brain centers
- c) Pain is generated only in peripheral nociceptors
- d) All nociceptive signals produce identical pain perception
- e) Pain perception can be altered by descending facilitatory & inhibitory mechanism.



THANK YOU
FOR YOUR
ATTENTION